AO1: Evaluate the significance of the interactions that sustain life and energy production by analysing the structure of cells and tissues, the roles of biomolecules (water, lipids, proteins, and nucleic acids), and the processes of ATP production, cell division, and protein synthesis, to apply these biological principles to challenges in genetic technologies and health improvement.

ITEM 1.1

Hospitals in Uganda have reported rising cases of heart and other related diseases raising tension in the public about the possible causes and risk factors. At a hospital in Kampala, Dr. Nkenga is studying why some adults develop heart problems even without common risk factors. One patient, Musa (32) shows symptoms such as irregular heart beat and weak heart muscles. Exposure to high levels of arsenic in body damages DNA sequence in region that codes for synthesis of actin and myosin leading to formation of non-functional fibrous proteins in cardiac muscle cells leading to poor contraction. High arsenic levels also affect and destroy DNA in region that controls formation of MICOs the complex proteins that control the folding of inner membrane of the mitochondria

Laboratory analysis of cellular and tissue changes revealed the information in Table 1

Table 1: Data table.

Parameter	Healthy heart tissue	Diseased heart tissue.
Arsenic Levels (ppm)	0.012	6.248
Gene that codes for actin in cardiac muscle cells	CCT, CAA,TCT,AAT,ACC,GAC	CCT,CAC, TCT,AAT,ACC,GCC
Fibrous tissue (Scar tissue)	Very little	Much (tissue becomes stiff)
Average number of Cristae on mitochondria	20.6	12.4
Mitochondrial oxygen consumption(nmol/min/mg)	55	30

Dr. Nkenga, after critical analysis of the above data, proposed three management interventions to help Musa

- Regular cardiac exercise

- Gene therapy
- being injected with oxygen-rich glucose intravenous supplements.

Task

(a) Analyse the effect of air pollution on cellular and tissue structures in Musa in relation to observed symptoms.
(b) Evaluate the effectiveness of Dr. Nkenga's management interventions in improving Musa's condition.

ITEM 1.2

Uganda is experiencing a steady increase in individuals with Type-I diabetes, a condition where body develops autoimmune reaction and mistakenly attacks and destroys insulin producing beta cells in the pancreas as a result glucose accumulates in blood causing high sugar levels. High glucose level narrow and harden blood vessels which reduces blood flow and delivery of oxygen and nutrients to cells. Insulin therapy is essential, yet hospitals struggle with high costs, irregular supply and patient complications linked to commonly use animal-derived insulin.
A 40-year-old man with type I diabetes for 10 years has recently developed a small cut on his foot that remained unhealed for several months despite receiving basic treatment; the man feels fatigued all time.
Laboratory investigations were conducted to compare selected cellular characteristics of his skin cells with those of a healthy individuals.
Table 1: Cellular characteristics of skin cells

Parameter	Healthy individual	Diabetic Patient
Cell membrane permeability to glucose (au)	100	68
Glucose uptake rate (micromole/cell hour)	8.5	3.4
Fibroblast mitotic index (%)	8.6	3.1
Time taken for the wound to heal	7days	30 days
Intracellular ATP concentration in foot cells (au)	100	38

The Ministry of Health advised the government to finance production of and use of recombinant insulin from bacteria to treat type I Diabetic patients instead of using insulin from animals.

Tasks

(a) Analyse observed differences healthy individual and diabetic patient.

(b) Evaluate the advice given to the government by the Ministry of Health. And propose other two justified interventions to help type I diabetic patients

ITEM 1.3

For more than 15 years, factories in an industrial area of Kampala have released untreated smoke containing heavy metal D that irreversibly binds to a channel protein that regulates water movement into the mucus in respiratory airways, hydrogen cyanide in nearby communities. Residents frequently reported persistent cough, fatigue, breathing difficulties. Doctors at the regional hospital observed that many factory workers developed lung cancer, liver damage and abnormal skin growths.

Medical researchers suspected that toxic chemicals were damaging DNA in body tissues such as lung epithelium, liver tissue and bone marrow.

They also believed that poisonous gases were interfering with aerobic respiration

Laboratory investigations showed reduced activity of DNA repair enzymes and respiratory enzymes in exposed workers, leading to accumulation of mutations and uncontrolled mitosis.

Table 1: Respiratory and Tissue Data.

Group	Rural control community	Factory-exposed workers
Mutation frequency in cell cycle control genes (%)	1	20
Amount of ATP in cells (au)	95	40

Viscosity of respiratory mucus (cP)	93	655
Rate of abnormal cell division (%)	2	25

Task

(a) Analyse data and explain how exposure to industrial pollutants resulted into above symptoms.
(b) Suggest and evaluate sustainable environmental, industrial and public health measures that could reduce respiratory diseases, tissue damage, and cancer risk in industrial communities of Uganda.

ITEM 1.4

A couple in Uganda had a boy child who showed delayed growth, poor muscle strength and 1month to heal from minor injuries while other children were taking a week to heal from minor injuries. Due to poor muscle strength, the boy struggled to walk and often fell.
One day, the boy fell and hit his lower abdomen against the edge of the table and urine leaked into the abdominal structures causing a urine-like smell from the abdomen
At the hospital, doctors suspected Leigh Syndrome, a mitochondrial disorder caused by a mutation in the SURF1 gene leading to mitochondrial complex IV (cytochrome oxidase) deficiency. The PCR and other laboratory investigations gave the following results:
Table 1

Test	Boy	Normal.
DNA sequence for SURF1 gene	5' CTC GAA AAA CTG 3'	5' CTC GAA GAA CTG 3'
Amino acid sequence	Leucine- Glycine-Lysine -Leucine	Leucine- Glycine-Glutamic acid- Leucine
Average ATP production per mitochondrion (%)	30	60
Rate of mitosis in damaged tissue cells	Low, few cells in metaphase	Normal rate of cell division

Transitional epithelium of bladder	Pores in basement membrane between some layers	No pores between layers.

Doctors confirmed Leigh Syndrome and leakage of urine into abdominal cavity. The doctor advised the couple to have the boy for gene therapy to eliminate the conditions in the boy.

Tasks

a) Analyse the data and explain how the mutation results in observed changes in the boy.

b) Evaluate the doctor's advice to the couple. And propose any other two justified strategies to help the boy

ITEM 1.5

John was persistently wheezing and feeling fatigue. He was then taken to a specialized hospital for medical care. Basing on his signs and symptoms, the doctor suspected a condition affecting the epithelial tissue lining John's respiratory system.
Normally, the cell membrane of the epithelial cells of the tissue contains a protein that regulates water movement in and out of the epithelial cell. The synthesis of the protein is controlled by a gene called cystic fibrosis transmembrane conductance regulator (CFTR) gene.
A defect in the CFTR gene can affect the protein function. Since the doctor suspected a defect in the protein, he requested for a CFTR gene test and other parameters.
The laboratory report from the doctor's request is shown in Table 1.
Table 1: Showing John's laboratory report

Parameter	John's results	Normal range
Sequence of CFTR gene	ATG, GCT, TTT, GAC, ACA	ATG, GCT, CTT, GAC, ACA
Rate of water movement out of epithelial cells	Low	High
Mitochondrial activity	Low	High
Thickness of mucus on epithelial cells	Very thick	Thin
Status of airways	Blocked	Open

The doctor recommended that John takes a supplement of a coenzyme for fatigue, then either gene therapy or dilators for persistent wheezing (dilators act on the muscle cells of the bronchioles making them relax and widen for some time).

Your classmates saw John's laboratory report and the doctor's recommendations and would wish to understand what they mean in relation to human health.

Task:
a) Analyse the laboratory report in relation to John's signs and symptoms.
b) Assess the effectiveness of the doctor's recommendation.

ITEM 1.6
The fish farmers in Kalangala District recorded reduced fish growth, poor survival and delayed recovery from injuries among the Nile tilapia reared near a metal-processing factory whose waste materials drain into the ponds. The fish exhibited weak swimming activity and slow wound healing. Water analysis revealed the presence of malonate, a known inhibitor of succinate dehydrogenase in the Kreb's cycle. The microscopic examination of fish tissues showed abnormalities in mitochondria.

Laboratory studies also detected reduced rate of fibroblast division and rate of transcription in affected fish.
Table 1: Cellular and physiological findings in Nile tilapia

Parameter	Fish in control ponds	Fish in polluted ponds
Succinate dehydrogenase activity (%)	100	34
Fibroblast mitotic index (%)	8.6	2.1
Average swimming speed	22	9
Blood pH	7.4	6.7

(a) Analyse the data and explain how drainage of waste materials into the pond caused the symptoms observed in the fish.

(b) Propose and evaluate sustainable interventions to increase fish productivity in the ponds.

ITEM 1.7

Mary, a 52-year-old woman, had been experiencing persistent fatigue, unexplained weight loss, loss of appetite and a painless lump in her left breast for four months. She later developed shortness of breath and persistent chest pain. She was referred to the Uganda Cancer Institute for specialised medical care. Based on her signs and symptoms, the doctor suspected breast cancer caused by a defect in a gene that regulates cell division.

Normally, body cells contain the TP53 gene, which controls the synthesis of the p53 protein. The p53 protein monitors DNA damage, repairs damaged cells or triggers programmed cell death (apoptosis) in cells that cannot be repaired. This prevents uncontrolled cell division and tumour formation. A mutation in the TP53 gene may alter the structure of the p53 protein, reducing its ability to control the cell cycle. As a result, damaged cells continue dividing rapidly, forming tumours that require large amounts of glucose and energy for continued growth.

To investigate the suspected mutation, the doctor requested a TP53 gene test together with tests on cellular respiration and tumour growth. The laboratory report is shown in Table 1.

Table 1: Mary's laboratory report

Parameter	Mary's results	Normal range
Sequence of TP53 gene	ATG, GCT, TTT, GAC, ACA	ATG, GCT, CTT, GAC, ACA
Average number of functional mitochondria per cell	28	12-18
ATP produced per glucose molecule	36-38	30-38
Cell division rate	Very high	Controlled
Tumour size	Large	No tumour

After reviewing the report, one Oncologist recommended gene therapy to restore the normal TP53 gene so that functional p53 protein could be produced. Another Oncologist recommended chemotherapy combined with drugs that reduce glucose uptake by tumour cells, arguing that limiting energy supply would slow cell division.

Your classmates have seen Mary's laboratory report and the doctors' recommendations and would like to understand how the results explain her condition and how they can be used to improve human health.

Task

a) Analyse the laboratory report in relation to Mary's signs and symptoms by explaining how the mutation in the TP53 gene affected the parameters above.

b) Assess the effectiveness of the two treatment recommendations.

Item 1.8

Two patients presented to a busy clinic in Kampala on the same day.
One was Mukasa, a 42-year-old non-smoking male, while the other one was Mutabazi, a 57-year-old male with a 12-year history of cigarette smoking of at least 10 cigarettes per day.
Both patients presented with shortness of breath and exercise intolerance (general body weakness). Further analysis showed that Mukasa had a hyper inflated barrel-shaped chest and no cough whereas Mutabazi had a normal shaped chest but had developed cough over the past five years.
Genetic analysis of both individuals revealed that Mukasa had a different sequence in the SERPINA gene, which is responsible for the synthesis of alpha-1-antitrypsin (AAT). (NOTE: alpha-1-antitrypsin (AAT) is a serum protease inhibitor that normally neutralizes neutrophil elastase, an enzyme released by white blood cells in the alveoli of lungs).
Mutabazi had normal findings on genetic screening. The findings from all the investigations were summarized in the report below.

Table 1: A report of findings from Mukasa and Mutabazi in comparison to a healthy individual*

Parameter	Mukasa (non-smoker)	Mutabazi (Smoker)	Healthy individual (non-smoker)
Triplet code at position 342 of DNA sense strand	AAG	GAG	GAG
Blood Serum AAT concentration(\mug/cm^3)	25.0	220.0	235.0
Relative mass of elastin tissue in alveoli (%)	32.4	56.8	98.5
Mitochondrial Oxygen consumption (nmol O_2/min/mg tissue)	41	55	100
Relative bronchial mucus production (%)	100 (normal)	285 (elevated)	100 (normal)

The attending physician had advised that both patients should be given direct oxygen therapy via oxygen masks.
However, when a senior pulmonologist was consulted, he recommended that Mukasa should immediately begin intravenous AAT augmentation therapy and Mutabazi should start on antioxidants as well as cease from smoking.

Task
(a) Critically analyse the findings above and explain the different signs and symptoms experienced by Mukasa and Mutabazi
(b) How effective will the doctor's advice and recommendations offered be to Mukasa and Mutabazi

ITEM 1.9

Farmers in Kigezi highlands have been depending on growing apples as part of their sources of income. Due to an increasing market demand for apples, the local agricultural cooperative in Kigezi introduced a new agrochemical intended to increase the size of the apples grown. The agrochemical was known to disrupt microtubule formation in plant cells. When the farmers sprayed the chemical directly onto very young apple fruitlets and buds of young apple trees in the nursery, the resultant apple fruits produced at the end of the season were indeed giant apples that grew to twice the normal size.
However, the farmers reported that while the skin of the apples looked healthy, the deep inner core tissues of the giant apples were rapidly rotting on the branches and emitting a distinct, sweet-alcoholic smell. Some of the farmers also noted that the treated apple trees took abnormally longer time for their apple fruits to reach full maturity.

To investigate the concern, a researcher analysed different properties of the cells from the outer skin and the deep inner core of the giant apples, comparing them to standard, untreated apples. The laboratory findings are presented in Table 1
Table 1: Cellular and Biochemical Analysis of Apple Tissues

Parameter	Stand Apple (untreated)	Giant Apple (Treated) Outer Skin	Giant Apple-(Treated)-Deep Core
Chromosome number per somatic cell	34	68	68
Average cell volume	4500	18000	18000
Surface area to Volume ratio of cells	High	Very low	Very low
Ethanol concentration in tissue (mg/g)	0.01	0.05	14.50
ATP yield per molecule of glucose	High	High	Very low

Duration of S-phase of the cell cycle of each somatic cell	Normal (low)	Elevated	Elevated

In order to avoid losses, the official agronomist from the cooperative suggested immediately spraying the trees with a chemical that increases the permeability of plant cell walls and membranes to gases.
A visiting botanist argued that the giant apples are genetically unstable and recommended abandoning the chemical completely and instead focus on selective breeding of standard apples.

Task
(a) Analyse the laboratory data and explain the different changes that occurred in the produced apples after introduction of the agrochemical.

(b) Evaluate the effectiveness and sustainability of each of the suggestions made by the agronomist and the botanist.

AO2: Evaluate plant structure and physiology by analysing structural adaptations and photosynthetic pathways in C3 and C4, environmental influences on photosynthesis, plant adaptations (to water availability), growth, photoperiodism, and the hormonal control of growth, to promote sustainable agricultural practices that improve crop yield and food security.

ITEM 2.1
Farmers in Uganda have faced unpredictable weather patterns that has resulted in poor yields in maize and beans leading to inadequate food to societies since they are some of major food crops and have given poor yields over seasons. So to harvest good yields farmers have been advised to grow these crops in green houses.
A farmer-training Centre in Uganda tried to investigate the conditions required in green houses to ensure productivity of the crops. They established two greenhouses A and B to investigate how photosynthetic pathway, environmental conditions and growth regulators affect crop productivity. The researchers grew maize (C_4) and bean (C_3) under different greenhouse conditions. They recorded the following parameters after six weeks.
Table 2.1

Parameter	Greenhouse A with Gibberellins	Greenhouse B with abscisic acid	Green house A with abscisic acid	Green house B with gibberellins
Temperature(ºc)	22	32	22	32
Photosynthetic rate in maize (relative units)	45	90	45	90
Photosynthetic rate in beans (relative units)	50	34	50	34
Percentage germination of maize seedlings	81	31	32	90
Percentage germination of bean seedlings	95	21	35	75

Tasks

(a) Analyze the crop productivity and germination of the two crops in the green houses.
(b) Propose justified ways how farmers can ensure high productivity of the crops in the green houses

ITEM 2.2

A commercial horticultural farm in Uganda produces spinach, chrysanthemums and other high-value ornamental flowers for local and export markets. The farm has experienced inconsistent flowering, excessive vegetative growth and reduced flower yield during periods when natural day length and environmental conditions change.

To improve production, the farm established a greenhouse to investigate how photoperiodic and plant hormones influence the transition from vegetative growth to flowering. Plants were exposed to controlled light–dark regimes while other growth conditions were maintained as uniformly as possible. The researchers recorded the following results after eight weeks.

Table 2.

Greenhouse	Light perio	Uninterrupted dark	Mean chrysanthem	Mean spinach	Flowering chrysanthem	Flowering spinac

treatment	d (h day^{-1})	period (h $night^{-1}$)	um shoot length (cm)	shoot length (cm)	um plants (%)	h plants (%)
A	10	14	28	50	88	15
B	12	12	35	40	72	37
C	14	10	42	35	38	69
D	16	8	48	27	15	90

They also observed that the chrysanthemums whose shoot apices were removed formed averagely 13 branches per plant and those that had intact shoot apices formed 5 branches per plant.

Task

(a) Analyse the greenhouse data to explain how light length influence the observed parameters on the plants.

(b) Propose justified ways how the maximum productivity in the two crops can be achieved.

ITEM 2.3

Farmers in Semuto Sub County experienced heavy weed invasion of *Heteropogon contortus ssp*, spear grass in fields planted with eggplant which is a major vegetable in their markets. The weed remained scare when the soil was left undisturbed, but after ploughing at the planting season, characterized with heavy rains which promote emerging of large number of weed seedlings that completed strongly with the egg plants. To control the weed, farmers applied a selective herbicide that inhibits electron transport in photosystem II in some plants. However, shortly after spraying many eggplant seedlings dried while the weed continued growing rapidly and soon dominated the field.

Agricultural scientists investigated the germination response and photosynthetic activity of both plants under the field conditions. Their findings are in the table below.

Parameter	Egg plant	Spear grass.
Bundle sheet malate concentration (mg/g)	0	60.4
Thylakoid electron transport rate (with herbicide)	Reduced	Normal

Rate of NADPH accumulation in chloroplast	Extremely low	High
Germination % in complete darkness	92%	5%
Germination % after red light exposure	94%	96%
Active phytochrome ratio (Pfr/Pr) required	0.10	0.85

The agricultural scientist however intends to formulate a new systemic herbicide that specifically inhibits malate synthesis as a way of eliminating weed.

Task:
(a) Analyse the differences in germination and photosynthetic responses between eggplant and spear grass under the field conditions.

(b) Evaluate the different farmer's interventions that would improve the survival of egg plants and whether the scientist's new formulation should be dropped or promoted.

ITEM 2.4.
Due to frequent changes in climatic conditions that lead to poor crop yields, farmers have been advised to start using greenhouses for tomato growing. A greenhouse is a house-like structure covered with a transparent/translucent roof and walls that create a controlled environment for growing crops.

To determine the suitable conditions to be used in the greenhouses, the farmers hired an agricultural officer who carried out trial tomato-growing using four greenhouses. Each greenhouse had different conditions and the results of the trial are as shown in table 2.4.1

Table 2.4.1: Showing tomato growing under different greenhouse conditions

Growth and physiological traits	Greenhouse and its conditions			
	Greenhouse 1 (32°C + auxins)	Greenhouse 2 (32°C + gibberellins)	Greenhouse 3 (25°C + auxins)	Greenhouse 4 (25°C + gibberellins)

Germination rate (%)	Moderate	Moderate	Moderate	high
Photosynthetic rate (µ mol $CO_2/m^2/s$)	High	High	low	low
Number of branches	Few	Very many	Few	many

For good yields, the agricultural officer recommended that the conditions of the greenhouse should *not* be kept constant all through. The farmers need to know how to implement the recommendation
Task:

a) Analyse the relationship between the conditions of the greenhouse and the growth and physiological traits of tomatoes.
b) Justify the recommendation of the agricultural officer and advise the farmers.

ITEM 2.5
In Rukungiri District, prolonged droughts, erratic rainfall, and increasing temperatures have greatly reduced crop productivity. To improve food security, a farmers' cooperative established a climate-controlled greenhouse to determine the most suitable environmental conditions for growing maize and beans throughout the year.
Extension workers grew the two crops under four different greenhouse conditions by varying temperature and carbon dioxide (CO_2) concentration. The results are shown in Table 2.5.1
Table: Growth and physiological performance of maize and bean Crops under different greenhouse conditions.
Table 2.5.1

Physiological trait	Greenhouse 1(20ºc + ambient CO_2)	Greenhouse 2(20ºc + High CO_2)	Greenhouse 3(35ºc + ambient CO_2)	Greenhouse 4(35ºc + High CO_2)
Beans				
Photosynthetic rate	Moderate	Very High	Low	High
Stomatal opening	Wide	Partially open	Wide	Partially open
Water use efficiency	Low	High	Low	Moderate

Biomass accumulation	Moderate	Moderate	Low	Moderate
Maize				
Photosynthetic rate	Moderate	Moderate	High	High
Stomatal opening	High	Slightly reduced	Slightly reduced	Reduced
Water use efficiency	Moderate	High	Very high	High
Biomass accumulation	Moderate	Moderate	High	High

The members of the farmers' cooperative would like to understand how temperature and carbon dioxide concentration influence the performance of maize and bean crops.

Task

(a) Analyse the data given in different greenhouse conditions and the growth and physiological performance of the maize and bean plants.

(b) Give justified ways on how greenhouse conditions should be managed to maximize the productivity of both maize and bean crops.

ITEM 2.6

A greenhouse Seed Company in eastern Uganda that supplies pearl millet (*Pennisetum glaucum*) seeds and beans to farmers was investigating how to improve seed yield and food security. Pearl millet (*Pennisetum glaucum*) is a C4 short-day plant while beans are C3 day neutral plants. Seeds harvested from each cycle were stored and used to establish the next crop. During the second cycle, failure of the blackout curtain interrupted the uninterrupted dark period.

Table 2.6.1

Production cycle		Germination (%)	Flowering (%)	Mean photosynthetic rate (relative units)	Seed yield (kg)
Cycle 1	Millet C4	95	96	85	548
	Beans C3	95	96	52	420

Cycle 2	Millet C4	94	37	62	169
	Beans C3	93	94	38	310
Cycle 3	millet C4	76	77	58	358
	Beans C3	68	95	35	245

Cycle 3 was established using seed harvested during Cycle 2 after the blackout system had been repaired.

Table 2.7.2

Component	Observation
Seed storage	Suitable throughout, at 15^0C, low humidity but abscisic acid level was high in cycle 2 seeds
Irrigation	Adequate in cycle 1, partly closed in cycle 2 and not provided in cycle 3
temperature	Maintained at 34 ^{0}C throughout

Two interventions have been proposed to improve yield

Grow beans under cycle 2 conditions

Task

(a) Analyse comparative performance of bean and pearl millet in the three growing cycles.

(b) Propose and evaluate the production system for maintaining commercial seed supplies, highlighting its strengths, limitations and suitability.

ITEM 2.7

A commercial tea farmer in Mukono District intercropped tea bushes with Napier grass along hillsides to check soil erosion. During the first few months, the system was highly effective. However, as the season progressed, the Napier grass grew tall and formed a dense canopy over the tea bushes.

The farmer soon noticed that the tea leaves turned pale green, and the bushes grew vertically with single, thin stems, producing very few lateral branches for plucking. Furthermore, during brief sunny intervals, the soil beneath the dense grass dried rapidly, causing the lower leaves of the tea bushes to drop off, while the Napier grass remained lush and vibrant.

An agricultural extension officer set up experimental plots under different management conditions to help the farmer optimize tea yields. The collected data is presented in Table 2.7.1

Table 2.7.1: Growth and physiological traits of tea and Napier grass

Parameter/ Trait	Plot A Open Sunlight (Uncut Grass)	Plot B 70% Shade by Napier Grass	Plot C Grass Cut Regularly + cytokinins sprayed on Tea	Plot D Open Sunlight+ Exogenous Auxins sprayed on Tea
Tea Net Assimilation Rate (umol CO: ms)	14.2	-1.5	13.8	14.0
Napier Grass Net Assimilation Rate (µmol COs ms)	32.8	8.1	12.4	31.8
Number of Lateral Bud Sprouts per Tea Bush	22	2	48	1
Tea Stem Elongation Rate (mm/week)	12.0	45.0	4.0	52.0

The agricultural extension officer recommended that the farmer should regularly slash the Napier grass to a height of less than 0.5 meters, mechanically clip the apical buds of the tea bushes, and apply a light organic mulch around the tea roots.

Task:

(a) Analyse how the varying conditions in the plots (As seen from table 2.8 above) contribute to changes in physiology and structure of the tea bushes.

(b) Justify the extension officer's recommendations and advise the farmer on how best these interventions will restore tea productivity.

ITEM 2.8

Residents of Nakasongola and Kotido districts have been pastoralists since ancient times surviving on rearing cows for meat ml and income. For cows to grow well they must be feed of pasture of both grasses and legumes. However prolonged dry spells in Nakasongola and Kotido districts have pushed daytime temperatures up to 35°C causing severe pasture scarcity for livestock. To address this crisis, the National Agricultural Research Organization (NARO) tested two drought-tolerant fodder species: Species A, a legume which a C3 plant and Species B a grass which is a C4 plant

Researchers collected the carbon dioxide exchange data as shown below in the table 2.8.1 during the trials.

table 2.8.1:

Temperature (^{0}c)	Net photosynthesis/mg CO: exchanged per dm^3 of leaf surface per hour		Photorespiration/mg CO_2, exchanged per dm^3 of leaf surface per hour	
	Species A (a C_3 plant)	Species B (a C_4 plant)	Species A (a C_3 plant)	Species B (a C_4 plant)
20	27	24	5	0
35	25	37	11	0

Task:

a) Analyse the relationship between net photosynthesis and rate of photorespiration of the two species at different temperatures
b) b) Devise justified ways farmers can use to be able to grow both crop species.

ITEM 2.9

As part of the government's program to increase production of medicines from natural tree extracts, National Drug Authority (NDA) established a herbaceous plant called Moringa plantation using seeds. The herbal medicine is etracted from leaves of this plant and for maximum benefit the tree should become bushy and develop many branches. The farmers realized that when the plant is sprayed with auxins the plant of average grows 3-5 branches per tree but if

treated with cytokinins during growth each tree bears 15-20 branches. However, when the seeds were sown, only a few germinated. After transplanting the germinated seedlings, they introduced a cover crop to conserve soil moisture and suppress weed growth. Unlike the Moringa trees, the cover crop uses two carbon dioxide fixing enzymes during photosynthesis. During the hot season, the plantation manager observed that the Moringa trees were not growing at the expected rate. To establish the causes of stunted growth, an agricultural officer conducted a study to compare the photosynthetic performance of Moringa trees and cover crops and also investigated the factors responsible for the low percentage germination of Moringa tree seeds

Table 2.9.1: Photosynthetic characteristics of the Moringa trees and cover crops

Plant	Activity of enzymes (%)		Soluble sugars produced (g plant day)
	Rubisco	PEP Carboxylase	
Moringa tree	34	Enzyme absent	89
Cover crop	70	90	159

Table 13: Hormonal changes associated with germination of moringa seeds.

Storage condition	Abscisic acid (µg g^{1} seed)	Gibberellin (ug g^{1} seed)	Amylase activity (%)	Germinating Seedlings (out of 100 seeds)
Cool, moist	5	20	86	91
Hot, dry	17	8	38	42

To improve seed germination and synthesis of medicinal products, the agricultural officer recommended the following interventions:

i) Treatment of all stored seeds with gibberellin immediately after harvesting.
ii) Planting Moringa n a greenhouse with enriched carbon dioxide
iii) Re-locate the Moringa tree plantation to a highland area

Task:
(a) Analyse the causes of the differences in sugar production and seed germination.
(b) Evaluate the effectiveness of the recommended interventions and advice the plantation manager on how to apply them.

AO3: Evaluate the structure and physiology of animal sensory organs and systems (circulatory, nervous, immune, and homeostatic) by analysing blood circulation, gas transport, immunity (vaccination and allergies), neural transmission, sensory perception, homeostatic control, and the role of adaptive behaviours in survival, to make informed health decisions and promote animal welfare

ITEM 3.1

An illegal night fisherman was moving along the shores of Kazinga channel in Queen Elizabeth National Park when he stepped onto a grassy area near the waters age. Moments later, a large territorial wild animal suddenly charged from a nearby thicket toward him. The fisherman immediately jumped backward, turned and ran at full speed across an uneven ground toward a rocky slope. Visibility was very poor due to darkness, and he had to navigate the unstable terrain while moving at high speed. During the escape, he fell and sustained a deep cut in his left thigh caused by sharp rocks. Despite this injury, he continued moving for several minutes before reaching a small cave where he remained lying down but now in excessive pain. About one hour later, the rescue teams arrived and took him to a nearby health post.

At the health post, the clinician injected a small volume of lignocaine drug a drug that blocks voltage gated channels in nerve cells around the area of injection around the injured area. Shortly afterwards, the affected region became completely insensitive to touch and pain, and the patient was unable to move his left leg properly for several hours. Physiological measurements were taken at different stages and these are shown in Table 3.1.1

Table 3.1.1: Physiological Data of the Fisherman.

Evaluated Parameter	**Steady State (Resting)**	**During escape**	**After injection**
Blood flow to skeletal muscles (cm^3/min)	1000	4800	950
Blood flow to digestive organs (cm^3/min)	1400	250	1350
Membrane Potential of Local Thigh Neurons (mV)	+30	+30	-70

Sinoatrial (SA) Node Firing Frequency (min^{-1})	70	148	82
Blood Adrenaline levels (ng/mL)	0.18	1.25	0.40

Task:

a) Analyse the above information to explain how the fisherman's symptoms came about.
b) Assess the effectiveness of the injection medicine that the fisherman received and propose other interventions that could improve the survival of the fishermen.

ITEM 3.2

A commercial dairy farm in Rukungiri district imported a herd of high-yielding Jersey cattle to improve milk production. Two weeks after their arrival, the animals were released onto pasture. Shortly afterwards, farm workers noticed large numbers of biting tsetse flies around the grazing fields.
The cattle became restless whenever the flies were present. They repeatedly flicked their tails, twitched the skin over their backs and sides, stamped their feet, and frequently moved up and down the pasture area to avoid the insects. Within a week, some animals gradually became weak, lost their appetite, and produced less milk. They also developed persistently high body temperatures (fevers) resulting in rapid breathing and opening the mouth wider than usual during day despite remaining at rest.
Laboratory examination of blood samples confirmed the presence of *Trypanosoma brucei* (parasites spread by adult tsetse flies) in the blood. The attending veterinary officer continued monitoring one affected cow and recorded the observations shown in Table 3.2.1.

Table 3.2.1: Behavioral and Physiological Indicators of an Infected Cow.

Indicator Monitored	**Pre-Infestation Baseline**	**Day 3 post-infection**	**Day 10 post-infection**	**Normal Reference Range**
Tail flicking	2	28	4	1-3

(times/minute)				
Core Body Temperature (°C)	38.5	39.2	41.5	38.0-39.3
Trypanosoma Parasite Density in Blood (cells/mL)	0	120	8400	0
Plasma Specific Antibody Concentration (AU)	0	15	190	0
Plasma ADH (Vasopressin) Levels (pg/mL)	1.5	3.8	8.9	1.0-3.0

The veterinary officer considered three management options: spraying the cattle with a synthetic insecticide spray, administering a high-dose anti-pyretic drug that stops fever and doubling the herd's water allocation.

Task:

(a) Analyse the above information carefully to explain how the Jersey Cattle were affected.

(b) Evaluate the effectiveness of each of the three veterinary management strategies proposed to restore the health and production of the dairy herd.

Item 3.3

A fish farmer in western Uganda noticed that fish in one of his ponds became less active, showed delayed responses to stimuli, and exhibited increased signs of disease after a nearby agro-processing facility discharged untreated wastewater into a stream supplying the pond.

A fisheries scientist investigated how changes in the aquatic environment affected the physiological functioning of the fish. Samples were collected from fish in an unaffected pond and the affected pond for further investigations.

Table 3.3.1: Physiological Observations of Fish from Two Ponds

Parameter assessed	**Fish from unaffected pond**	**Fish from affected pond**
Dissolved oxygen in water (mg L^{-1})	7.8	2.6
Blood oxygen saturation (%)	94	61
Heart rate (beats min^{-1})	62	108
Response time to external stimulus (s)	0.4	1.5
White blood cell count ($\times 10^6$ mL^{-1})	8.2	17.4

Further investigation showed that the waste water reduced oxygen availability in the pond and introduced substances that stressed the fish. The scientist explained that survival in changing environments depends on effective coordination between respiratory, circulatory, nervous, and immune systems.

Task:

(a) Analyse how the environmental changes affected oxygen transport, circulatory regulation, nervous coordination and immune responses in fish from the affected pond.

(b) Propose sustainable strategies that fish farmers, local communities and industries could implement to prevent water pollution and physiological disruptions in aquatic organisms and improve aquatic animal health, productivity and welfare. For each strategy, justify how it contributes to these outcomes.

ITEM 3.4

A poultry farmer in eastern Uganda introduced a new housing system intended to improve egg production. During a period of unusually high environmental temperatures, the farmer observed that some birds became weak, showed reduced feed intake, produced fewer eggs, and developed frequent infections.

A veterinary officer investigated the physiological responses of two groups of birds kept under different management conditions.

Table 4: Physiological Observations in Two Groups of Birds

Parameter assessed	**Group A (Well adapted)**	**Group B (Affected)**

Core body temperature (°C)	41.0	43.5
Plasma ADH concentration (pg mL^{-1})	28.4	9.7
Urine osmotic concentration (mOsm kg^{-1})	1240	520
Antibody concentration after vaccination (arbitrary units)	86	34
Disease incidence during observation period (%)	8	41

Further investigations showed that Group B inadequate drinking water and poor management practices during periods of high environmental temperature. The veterinary officer explained that maintaining survival during heat stress requires coordination between the nervous system, endocrine system, kidneys, circulatory system and immune system.

Task

a) Analyse how disruption of physiological regulation affected temperature control, water balance and immune function in group B birds.
b) Propose sustainable management strategies that poultry farmers could implement to improve productivity on the farm.

ITEM 3.5

During a violent night storm on Lake Victoria, a fishing boat capsized, throwing two fishermen into the freezing water. Okello, 22-year-old managed to cling to a floating wooden crate and was rescued by a nearby patrol boat after approximately two hours in the water. Kasasa, a 55-year-old fisherman, was swept further by strong currents and could not be located immediately due to poor visibility and fluctuating water levels. He was eventually found nearly 3 hours later, drifting unconscious near a cluster of papyrus reeds.

Both men were rushed to a medical facility in Entebbe for emergency care. On arrival, Okello was conscious and shivering violently, with cold but stable extremities. In contrast, Kasasa stopped shivering entirely, was semi-conscious, profoundly cold, and was in shock (severe circulatory collapse). The emergency team immediately began assessment of their vital signs and homeostatic parameters upon admission. The findings are summarized below.

Parameter	Okello (rescued after 2hours)	Kasasa (rescued after 3 hours)	Normal range
Core body temperature(^oC)	34.2	29.5	36.6-37.5
Shivering rate	Intense	Absent	0 (at rest)
Mean arterial blood pressure()mmHg)	130/85	70/40	120/80
Blood oxygen saturation (%)	96	84	95-100
Blood glucose level(mg/100ml)	76	55	90-110

In an attempt to help Kasasa, the emergency team administered immediate intravenous isotonic solutions, transferred him to a warmer (at 32^0C), and offered a strong sucrose solution intravenously.

Task

a) Analyse how the freezing water differently affected the fishermen
b) Evaluate the emergency team's interventions in helping improve Kasasa's condition.

ITEM 3.6

Amina, a 17-year-old student from Gulu, had always been among the top runners in her school. Despite occasional episodes of joint pain and fatigue, she had never been formally diagnosed with any illness. During the national cross-country championship held in Jinja, she performed excellently for the first 4 km. However, as the race continued and the heat intensified, Amina suddenly slowed, stumbled, and collapsed near the finish line, gasping for breath and clutching her legs in severe pain. Spectators noticed her fingernails and lips had turned bluish. Attending medical officers found her to be in acute pain crisis. She was evacuated to a Regional Referral Hospital where investigations were immediately conducted. Doctors discovered that her blood appeared dark and did not flow normally through micro filters used during blood typing.

Genetic analysis of her β-globin gene revealed the mRNA codon GUG at Position 6. The results from all other investigations done were compared with a healthy age-matched peer, are presented in Table 1 below.

Table 1:

Parameter	**Amina (after race)**	**Normal Range (Healthy Peer)**

Blood Oxygen Saturation	72	95-100
Muscle Cell ATP(µmol/g)	2.0	7.8
Plasma PH	7.28	7.40
Erythrocyte cytoskeleton integrity (spectrin-actin network).	Rigid	Flexible

The treating physicians have proposed to administer high-flow supplemental oxygen to Amina immediately, provide intravenous fluids with glucose and they intend to prohibit Amina from engaging in any more athletics.

Task

(a) Analyses the above information and explain how Amina's symptoms came about.

(b) Evaluate each of the proposed interventions by the physicians.

ITEM 3.7

After attending a 6-hour outdoor music festival featuring extreme noise levels, dancing and consuming large quantities of alcohol, an 18-year-old student, David, was admitted to the school's sickbay. He presented with inability to walk in a straight line, profound dehydration and complaints of a constant "ringing" in his ears.

The university nurse assessed David and compared his vitals to a healthy, sober student who did not attend the concert.

Table 3.7.1

Parameter	David (Post-Concert & Alcohol)	Healthy Student
ADH Concentration in Blood (pg/mL)	0.8	4.5
Cardiac Output (L/min)	3.2	5.1

Sensory Hair Cell Integrity in Cochlea	35% damaged	0% damaged
Endolymph fluid dynamics in Semicircular Canals	Highly turbulent / Altered viscosity	Normal flow

Following his admission, the medical team proposed to give him isotonic saline solution intravenously, oral rehydration salts and place him in a quiet rehabilitation environment. The attending doctor however later advised that David should quit alcohol to avoid long term complications.

Tasks

a) Enalyse how David's parameters varied from those of a healthy student and how this led to the David's presentation.
b) Evaluate the medical team's proposal in helping David and support the attending Doctor's advice.

ITEM 3.8

During a Uganda People's Defense Forces (UPDF) recruitment exercise in Kamuli District, Eastern Uganda, shortlisted candidates were required to complete a 15KM endurance run as part of a physical fitness assessment. The exercise was conducted on a very hot afternoon with an ambient temperature of about 32.7°C and high humidity. Mark, a 23-year-old candidate, maintained a fast pace during most of the race. However, about 250 meters from the finish line, he became confused, staggered, and suddenly collapsed. Witnesses reported that Mark had been sweating heavily during first part of the race but later stopped sweating. They also observed him repeatedly pouring cold water over his head and body without drinking enough water.

The medical team immediately examined him and recorded the following.

Table 6: showing medical findings

Parameter	**Healthy hydrated individual**	**Mark**
Plasma ADH concentration	Low	High
Core body temperature (°C)	37.0	38.6
Heart rate (beat/min)	60-100	168

Blood pressure (mmHg)	120/80	90/60
Urine colour	Pale yellow	Dark yellow

Task

(a)Using the information provided, analyses the physiological changes that occurred in Mark during the marathon.

(b)Recommend and evaluate immediate first aid and medical interventions that should be used to improve Mark's condition at the scene and in hospital.

AO4: Evaluate the principles of genetics, evolutionary mechanisms, and ecological interactions by analysing Mendelian and non-Mendelian genetics, species evolution, speciation, resistance, extinction, population dynamics, ecosystem balance, and carbon emissions, to create sustainable strategies for managing invasive species, enhancing food security, and mitigating climate change.

ITEM 4.1

Communities living Mitooma district have been growing crops for many years harvesting high yields for sale and domestic consumption. However they have recently reported declining crop yields, increasing temperatures and rapid spread of invasive plant species. Environmental researchers from National Environment Management Authority and Uganda Wildlife Authority found out that the communities have been using a broad spectrum pesticide for the past 10 seasons control pests in their crops and no known enzyme in animal kingdom can digest this pesticide and the pesticide cannot be easily excreted in the kidneys and is highly fat soluble. They noticed that a large part of forest in Kiyanga a sub county in the district was cleared and an industry that processes coffee using energy generated from burning of petroleum products was established. They carried out investigations to study species evolution, resistance, extinction, population dynamics, and ecosystem balance. The researchers compared populations of insects, birds and carbon dioxide levels over a period of ten years to determine how environmental changes and human activities influence survival. Table 4.1.1 shows the results of the study. The researchers suggested using mass afforestation using fast growing exotic eucalyptus to restore biodiversity, complete ban of broad spectrum to reduce dearth of predatory birds and wiping out all invasive plant species using a narrow spectrum herbicide

Table 4.1.1: **Changes in populations of organisms in an ecosystem**

Organism/feature	Population in 2014	Population in 2024
Insect pollinators	9200	3400
Predatory birds	820	250
Percentage of resistant insects to the pesticide (%)	8	76
Percentage of susceptible insects to the pesticide (%)	92	24
Carbon dioxide levels in ppm	398	426.8

a) As an intern biologist working with NEMA, analyze these changes to the community and district of Mitooma.
b) Evaluate the researchers' suggestions to the communities.

ITEM 4.2

Farmers and agricultural extension workers in Kasese District and Mayuge District observed increased crop losses, declining fish populations, spread of invasive water hyacinth, and changing weather patterns. Due to their outcry government tasked National Agricultural Research Organization and National Environment Management Authority to study the situation and come out with causes of all these problems and possible solution to avoid famine. The two government agencies carried out a comprehensive research for a period of four years and came up with the following observations.
In a wetland ecosystem affected by invasive species there was dense water hyacinth cover, reduced dissolved oxygen in water and declining fish populations while in the farmland ecosystem there was reduced pollinator numbers, increased soil erosion and lower crop productivity

Table 4.2.1: Results from study on resistant bacteria, crop yield, native fish species and bush burning in a wetland and farmland ecosystem.

Year	Percentage tolerant bacteria to anoxic waters (%)	Crop yield (tons/year)	Native fish species	Average environmental temperature in degrees Celsius
2019	12	320	8500	21.8
2020	28	267	6720	22.0

2021	49	212	3900	23.0
2022	73	170	2100	24.1

In order to solve these challenges, the government has proposed mass spraying of water hyacinth using a broad spectrum herbicide to control their spread, plant fast growing exotic pine trees to absorb carbon dioxide in the atmosphere and promote biodiversity

Task

a) As a scientist explain to farmers and extension workers causes of these trends in the two districts.

b) Evaluate the government's proposals in controlling invasive water hyacinth, reducing carbon dioxide and promoting biodiversity in the two districts.

ITEM 4.3

An island is a piece of land completely surrounded by water. A large island in the Indian Ocean with great biodiversity of native birds and good crop yiields experienced the introduction of an invasive predatory snake species through international trade. The snakes greatly preyed on native birds and small mammals. Within 20 years, ecologists on this island observed changes in populations of native and invasive species and increasing economic losses due to crop damage caused by increasing population of insect pests as the population of insect eating birds declined.

Table 4.3.1

Parameter	Year 0	Year 10	Year 20
Native bird population	12000	7000	1000
Invasive snake population	50	3000	9500
Crop yield	25000	15000	3000

After 20 years farmers decided to spray the insect pests with a narrow spectrum pesticide consecutively for next 5 years which resulted to dearth of many insects. These insects lay eggs and their larvae grow up in hard tree logs. The percentage of native birds with long and hard beaks and those with short and soft beaks were recorded immediately and after 5 years and recorded in **table 4.3.2**

Table 4.3.2

Parameter	Year 20	Year 25
Percentage Native birds with long and strong beaks	20	75
Percentage native birds with short and soft beaks	80	15
Percentage dearth of pests after pesticide application	90	30

To control the invasive snake species, the government proposed killing the snakes with poisoned rat baits, or to import a large population of mongoose into the island the government also proposed clearing of the land to plant a fast growing genetically modified crop having a gene for pest resistance to the insect pests.

Task

(a) Analyze the data in table 5 and explain the observations of ecologists in this study.

(b) Suggest justified strategies that can be followed by the conservation agencies to promote diversity and restore ecosystem stability

ITEM 4.4

Ecologists in Mbarara district investigated the long term effects of pesticide use in controlling pests that destroy flowers of coffee plants in three coffee plantations separated by wide rivers. They observed that over the past seven years, the pesticide has become less effective as the beetle population has continued to increase to greater levels. They also found that some beetles possess pesticide resistant allele R, which allows them to survive repeated pesticide applications and this made resistant beetles to become more common over several generations.

At the same time, the decrease in population of susceptible beetles reduced food availability for insect eating birds and reduced pollination by insects. In the three isolated coffee plantations separated by wide rivers, researchers observed that the beetles were becoming genetically different, raising concerns that they could eventually form new species greatly resistant to all pesticide applications. The researchers concluded that:

(i) continued pesticide use will eventually cause resistant beetle populations to evolve into separate species

(ii) Use of pesticides is the most effective method of controlling pests.

Their results are summarized in table 4.4.1

Table 4.4.1

Year	Resistant beetles (%)	Susceptible beetles (%)	Number of insect eating birds	Number of pollinating insects
1	10	90	130	4000
2	35	65	110	3200
3	50	50	95	2500
4	68	32	80	1500
5	82	18	30	800

Ecologists remained puzzled as to what could be the reasons behind the observed trends in table 6
Task
(a) As a biology student explain the information given in the scenario
(b) Evaluate the conclusions of the ecologists in this investigation

ITEM 4.5
In Bushenyi District, vegetable farmers have reported decline in fruit quality and yield due to a rapidly spreading insect pest that damages leaves, stems and also spreads viral infections in the crop.
The main challenge is that some plants show pest resistance but produce very small fruits, while others have large fruits but are easily attacked by pests. This has made it difficult for farmers to select planting materials that combine both useful traits. They noticed that.

- A few plants showed both resistance and large size, but these were rare.
- Farmers under continuous mono cropping recorded more pest damage.
- Fields near edges and mixed vegetation had fewer insect pests.
  Some growers who used intercropping reported lower pest spread and more stable harvests.
- Pest attack was highest during the dry period when alternative host plants was scarce.

A research conducted by officers from the National Agricultural Research Organization (NARO) revealed that.

- Pest resistance is controlled by gene R, where R gives resistance and r gives pest susceptibility.

- Fruit size is controlled by gene S, where S gives small fruits and s gives large size.

A pure breeding resistant plant with small fruit was crossed with pure breeding susceptible plant with large fruit produced only resistant small sized fruits in F1. When F1 plants were testcrossed, the following off springs were obtained.
Resistant, small fruit 460
Resistant, large fruit 63
Susceptible, small fruit 62
Susceptible, large fruit 456
The findings together with field observations on pest population were discussed to increase crop management.
Tasks
(a) Analyze the findings made and how resistant crops with large fruit came about and how farm management practices contributed to pest increase and reduced crop yields.
(b) Propose and evaluate sustainable strategies to improve crop yield, quality and pest control.

ITEM 4.6.
Kalinzu forest is a large piece of land covered by a rain forest reserved by the government in in Rubirizi district having great biodiversity and attracting tourists around the globe. However,
Communities living around Kalinzu forest reserve in Rubirizi District have reported an increase in rodent damage to crops such as maize, cassava and millet. Following these complaints the National Forestry Authority NFA, conducted an ecological survey.
The investigation revealed that the population of native rodents species within the forest has declined, while that of non-native rodent species has increased significantly.
The non-native species possesses stronger limbs, larger incisors, a higher reproductive rate, and is more efficient at exploiting food reserves. As a result, the native rodents are displaced from the forest and increasingly invade nearby crop gardens.

The survey further established that illegal logging, charcoal burning, road construction and agricultural expansion have fragmented the forest into three

isolated patches A, B and C, consequently, movement of native rodents between patches has been restricted, limiting gene flow and interbreeding. Breeding experiments among native rodents from the 3 patches produced the following results table

Table 4.6.1: showing products of crossing rodents of different patches

Cross	Observation
A X B	No mating occurs
B X C	Mating occurs but the offspring are infertile
A X C	Mating occurs but fertilization does not occur because a zygote is never formed

(a) Using evidence from case study, analyse the ecological and evolutionary interactions responsible for the changes in rodent population in the forest reserve.

(b) Propose and evaluate a management strategies for forest reserve that restores the forest, controls the invasive rodents and conserves native biodiversity while supporting local livelihoods.

ITEM 4.7

In the semi-arid rangelands of Kazo District, pastoralists keep indigenous Zebu cattle adapted to harsh environmental conditions. Heat tolerance and water-retention ability in these cattle are controlled by a multiple allele system (T^H, T^M, T^S), where T^H (extreme tolerance) is dominant over T^M (moderate tolerance), and both are dominant over T^S (sensitivity). However increasing heard sizes has led to severe overgrazing causing a shift n vegetation from nutritious perennial grasses (*Themeda triandra*) to invasive, unpalatable weeds (*Sporobolus pyramidalis*). Ecologists have carried out several investigations on the cattle genetics and survival features and comparisons have been done on the land affected by overgrazing. The findings are summarized in tables 4.7.1 and 4.7.2 below.

Table 4.7.1 Cattle Genotype Frequencies and Survival Metrics during Extreme Drought

Cattle genotype	Population Frequency	Mean Daily Water Intake (L)	Mean Daily Water Intake (L)

T^HT^H	0.05	12	4
T^HT^S	0.15	14	6
T^MT^M	0.30	25	15
T^MT^S	0.40	28	19
T^ST^S	0.10	45	35

Table 4.7.2: Rangeland Ecological Succession Metrics across Grazing Gradients

Grazing Pressure Zone	Dominant Plant Species	Net Primary Productivity ($g/m^2/yr$)	Soil Erosion Rate (tons/ha/yr)
Light Grazing	Themeda triandra (Perennial)	500	0.5
Moderate Grazing	Mixed grasses & herbs	200	4.2
Heavy Overgrazing	*Sporobolus pyramidalis* (Weed)	110	38.0

As part of the ecologists' implementation strategies, they intend to roll out a mass culling program targeting to remove any cattle showing traits of the T^ST^S genotype to eliminate dearth of cattle during drought, dividing the grazing land into paddocks for rotational grazing to reduce grazing pressure and clearing the *Sporobolus pyramidalis* weed using broad-spectrum herbicides to allow native grasses to grow back.

Tasks

(a) Using Table 4.7.1, explain the inheritance pattern of heat tolerance in cattle and the selective pressures acting on each genotype and how varying grazing pressure affected the ecosystem using data from Table 4.7.2

(b) Evaluate the long-term effectiveness of the ecologists' implementation strategies

ITEM 4.8

The diamondback moth, *Plutella xylostella*, is a major pest of cabbage crops in tropical agricultural regions. Rising global temperatures and the widespread use of synthetic pesticides have increased the challenge of controlling this pest. Scientists investigated the inheritance of pesticide resistance and wing morphology under different environmental conditions.

Two inherited genes were studied:

Locus R/r Pesticide resistance (R- Resistant, r-Susceptible)

Locus W/w: Wing morphology (W-Long wings, w = Wrinkled wings)

A true-breeding resistant, long-winged moth (RRWW) was crossed with a true-breeding susceptible, wrinkled-winged moth (rrww). All F_1 offspring were resistant and long-winged with genotype RrWw.

To investigate whether environmental temperature affects patterns of inheritance, adult F_1 moths were divided into two groups and acclimatized to different temperatures before each group was test-crossed with homozygous recessive moths (rrww).

Condition A: Test cross conducted at 22°C (baseline temperature).

Condition B: Test cross conducted at 32°C, simulating prolonged global warming.

Table 4.8.1: Phenotypic distribution of test-cross offspring

Phenotype	Condition A (22°C)	Condition B (32°C)
Resistant, long-winged	470	500
Susceptible, wrinkled-winged	468	500
Resistant, wrinkled-winged	32	0
Susceptible, long-winged	30	0

Over a 10-year period, farmers repeatedly sprayed a synthetic chemical pesticide. Ecologists monitored the frequency of resistant allele (R) in the moth population and the population size of a native predatory beetle that naturally controls the moth population

Table4.8.2: 10-Year trends in resistance, predator population, and ecosystem health

Year	Frequency of resistant allele (R)	Predator population (per km²)
1	0.05	1200
4	0.22	650
7	0.58	110
10	0.95	0

Ecologists have proposed the following strategies to control the pest pest population

(i) Apply highly selective pesticides to kill moth pests

(ii) Introduce and conserve predatory beetles to control the moth pest

(iii) Practice crop rotation

Tasks:

a) Analyse how increased temperature and continuous pesticide application have affected the inheritance patterns of the moth, frequency of resistant allele and predator population.

b) Evaluate the effectiveness of each strategy in increasing long term crop productivity.

ITEM 4.9

Lake Victoria is shared by Uganda, Kenya, and Tanzania. It was once famous for its incredible biodiversity, holding over 500 unique species of cichlid fish that evolved from a common ancestor through adaptive radiation. In the 1950s, the predatory Nile perch *Lates niloticus* was introduced into the lake. Soon after, the invasive water hyacinth weed (*Eichhornia crassipes*) began spreading rapidly across the water surface. These changes completely altered the lake environment, creating a powerful selective pressure on the native fish populations. Ecologists and geneticists monitored the lake over several decades to study how these pressures affected both the ecosystem and the genetic traits of the surviving cichlid populations.

The key data collected are presented in the table 4.9.1 below:

Year	Nile perch population	Water hyacinth cover (%)	Number of active cichlid species	Frequency of G-gene (controls

				larger gill surface area in cichlid)
1945(before introduction	0	0	500+	0.05
1960	4000	0	380	0.06
1980	200,000	5	200	0.28
2026	350,000	15	150	0.82

The conservation team has proposed two strategy plans to be taken to restore the cichlid population.
(i) Capture a representative part of the remaining cichlid species and breed them in a controlled highly oxygenated and Nile perch free environment and through selective breeding further increase frequency of the G-gene, spread the water hyacinth with herbicides and extract as much Nile perch from the lake as possible.

(ii) Capture a representative part of the remaining cichlid species and breed them in a controlled highly oxygenated and Nile perch free environment and through selective breeding increase the frequency of the ancestral g-gene, introduce specific biological agents in the lake to reduce water hyacinth and also regulate Nile perch population.

Task
a) Analyse the information given in the table and the scenario to explain the changes that occurred
b) Evaluate the effectiveness of each strategy proposed by the conservation team, select and justify the most sustainable strategy

ITEM 4.10
Lake Bunyonyi, a deep crater lake in South-western Uganda, is home to a unique, endemic species of aquatic frog. Historical ecological surveys showed considerable variation in size of the eggs laid by the frogs within the population.
Over the last twelve years. There has been increased cultivation on the surrounding hillsides by local farmers who plough the land regularly to grow cassava for food. Ecologists have now started reporting an increasing change in

the lake's water quality. Studies conducted within Lake Bunyonyi over the past 12 years were summarised in the table 4.10.1below.

Table 4.10.1 Limnological Changes in Lake Bunyonyi Over 12 Years

Year	Algal Biomass (g/m^3)	Average Dissolved Oxygen (mg/l)	Frog Population (Adults per 100 m shoreline)
1996	12	8.4	450
2000	45	5.2	310
2004	110	2.1	145
2008	195	0.8	65

The local environmental officer has been consulted by the ecologist on how best the ecosystem and ecosystem of Lake Bunyonyi can be improved.

Task:

(a) Analyse the changes in algae biomass and average dissolved oxygen within Lake Bunyonyi and frog population over a 12 year period

(b) Propose justified ways to improve the structure of the lake's ecosystem.